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Low earth, high stakes: The LEO satellite race between AI demand and geopolitical fragmentation

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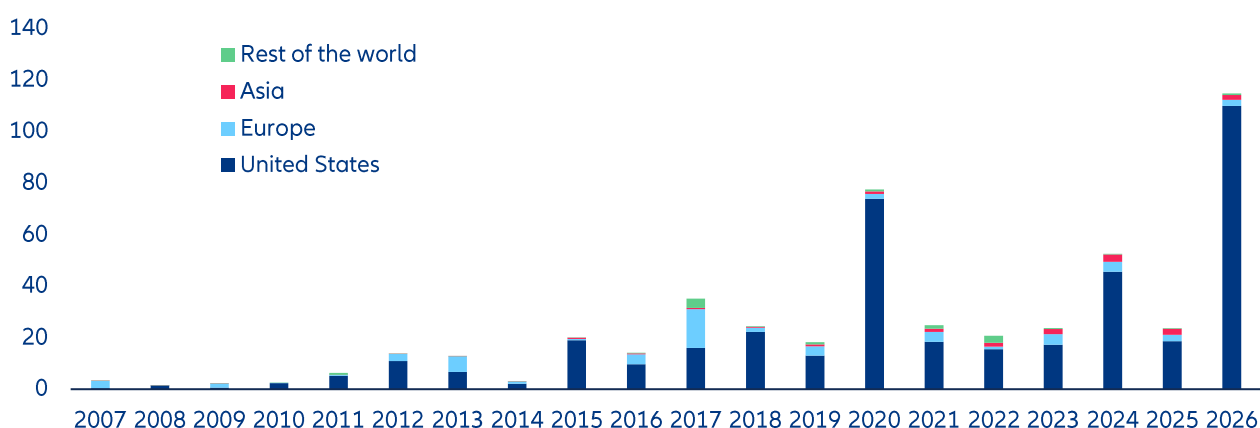
In Summary

- **SpaceX's record-breaking IPO puts a public price on a conviction that private investors already hold: space is a durable investment vertical, not a passing cycle.** Private investment in space tech hit a record USD12.4bn in 2025, up 48% year-on-year, even as venture funding cooled elsewhere, with the US capturing 60% (about USD7.3bn). Late-stage deals reached 41.3% of space-tech VC transactions, signaling capital concentrating on proven, revenue-generating businesses rather than speculative bets.
- **The next wave of the AI economy requires real-time connectivity that terrestrial networks cannot deliver. LEO satellites are the only viable solution.** Autonomous vehicles, industrial robotics, drone logistics and AI-driven grid management all depend on continuous, low-latency connectivity, which 5G cannot meet since 40% of global land surface sits outside reliable coverage.
- **The LEO services market is on track for a 7-8x expansion this decade, within a broader space economy set to triple.** Valued at about USD16bn in 2025, LEO services are projected to reach USD120bn by 2030, driven primarily by enterprise and industrial demand. The broader space economy, currently accounting for about USD600bn, is projected to exceed USD1.8trn by 2035.
- **The US holds a commanding lead through SpaceX's vertical integration and Amazon's hyperscaler logic, but China is closing the gap with state capital and sovereign ambition.** SpaceX's over 9,000 active satellites in orbit and its successful IPO position it as the default connectivity backbone of the AI economy; Amazon has invested over USD10bn in Project Kuiper targeting the same enterprise segment. Meanwhile, China's credible near-term pipeline stands at about 28,000 satellites across state-backed operators – expected to represent close to 40% of effective LEO satellites by 2030 – replicating the model used in EVs, with state funding eliminating the capital constraints that limit commercial rivals.
- **Emerging markets are the strategic battleground where the long-term balance of digital infrastructure power will be decided.** Fixed broadband penetration remains below 40% in Africa and 80% in Asia-Pacific and Latin America (vs. over 90% in Europe and North America). China has moved earlier and more deliberately than Western competitors, positioning LEO as the connectivity layer of Belt and Road's digital extension and replicating the Huawei playbook – below-market pricing, technical lock-in and conversion of infrastructure dependency into geopolitical leverage. India, Brazil and Gulf sovereigns have enough scale or strategic assets to negotiate terms; the remaining 150-plus economies face straightforward dependence.
- **Europe's best strategy is to own a chokepoint in the value chain rather than compete on constellation scale – the ASML model applied to launch.** With Eutelsat OneWeb's about 650 satellites financially constrained and IRIS²'s 280-satellite sovereign constellation limited by design, Europe cannot match the US-China pipeline of over 100,000 satellites. The credible path is deepening Arianespace's position as an indispensable launch provider – already validated by Amazon's Kuiper contract on Ariane 6 – while closing the reusability gap, building toward Ariane Next in the 2030s.

Investors are buying into the space race

Follow the capital, not the rhetoric. Where private money flows, and how much of it, is the clearest evidence that space has become a durable investment vertical rather than a passing enthusiasm. Private investment in space tech reached a record USD12.4bn in 2025, up +48% on the prior year and past the previous peak set in 2021, even as venture funding cooled across most other sectors. The money is also concentrating geographically with the US capturing roughly 60% of the global total, around USD7.3bn, propelled by launch and defense programs including the Pentagon's Golden Dome; Asia, led by China, added some USD2bn and Europe grew from a far smaller base, still lagging both. That concentration is the story the chart below tells; appetite is real but not evenly distributed, and the gap between the leaders and everyone else is widening.

Figure 1: Capital invested in space tech by global region (EUR bn)



Sources: Pitchbook, Allianz Research

Composition over headline: the deal mix confirms this paper's thesis. Commercial launch funding has eased as the segment matured into commoditized infrastructure that no longer commands venture multiples, while satellite communications held firm and geospatial intelligence rose. Investors are paying for the part of the stack closest to recurring revenue – connectivity and data – rather than the hardware that merely reaches orbit.

A maturing market, not a retreating one. Late-stage deals reached 41.3% of space tech VC transactions in 2025, up from 39% in 2024 and the highest share in a decade, a sign that capital is flowing to proven, revenue generating businesses rather than speculative bets. Exits are reviving too, primarily through acquisition as incumbents and hyperscalers buy capabilities cheaper than they can build them, the signature of an industry consolidating around a few winners rather than one still searching for a model. The fundamentals underwrite the conviction, the satellite communications market alone was worth roughly USD25bn in 2025, with LEO deployment, 5G integration and IoT as its named growth drivers.

The Starlink IPO is not an isolated event, but the public-market confirmation of a capital base that has already concluded that satellite connectivity is the segment with the clearest path to durable revenue. A listing of that scale would not open the door to broad-scale, massive space investing, but the capital is already flowing so much as to put a public price on a conviction that private investors reached some time ago.

Terrestrial networks cannot deliver the AI economy

The real-data economy is outgrowing terrestrial infrastructure. The first wave of the AI economy ran on data that already existed – text, images and historical records fed into training pipelines. The next wave is different. Autonomous vehicles navigating live traffic, industrial robots coordinating across factory floors, drone logistics

networks, precision surgical systems and AI-driven grid management all share a single dependency: continuous, low-latency, high-bandwidth connectivity to real-time data, at any location, with no tolerance for coverage gaps. This is precisely what current terrestrial networks cannot provide in a reliable way. 5G offers low latency and high throughput, but only where infrastructure has been deployed (i.e. dense urban zones in wealthy economies). Roughly 40% of the global land surface, including most of the oceans, remains outside reliable 5G coverage. For applications that move (e.g. vehicles, drones, ships, aircraft etc.) this gap is not a minor inconvenience but a significant operational constraint.

Satellite connectivity is not new, but the economics and physics of the previous generation made it unsuitable for real-time applications. Geostationary satellites orbit at approximately 36,000 km, introducing round-trip latency of 500-600 milliseconds – workable for broadcast and slow data transfer but entirely unsuitable for autonomous systems requiring sub-20ms response times. Low Earth Orbit (LEO) changes this equation fundamentally: at 550-1,200 km altitude, LEO constellations achieve latencies of 20-50ms, comparable to terrestrial broadband, while providing global surface coverage through overlapping orbital shells. The technological shift from GEO to LEO is therefore the architectural precondition for deploying AI-intensive applications at global scale.

Table 1: Technical differences between LEO and geostationary satellites

Attribute	LEO (low Earth orbit)	GEO (geostationary)
Altitude	~160–2,000 km (Starlink ~340–570 km)	~35,786 km, fixed over the equator
Round-trip latency	~20–50 ms (fiber-like)	~500–600+ ms
Satellites for global coverage	Hundreds to thousands (mega-constellation)	~3 cover most of the globe (poles excepted)
Satellite unit	Small, mass-produced, low unit cost	Large, bespoke, expensive
Footprint per satellite	Small	Very large (~1/3 of Earth)
Operational life	~5–7 yrs	~15 yrs
Terminal & link	Lower path loss → smaller, cheaper terminals	Higher path loss → larger antennas/power

Source: Allianz Research

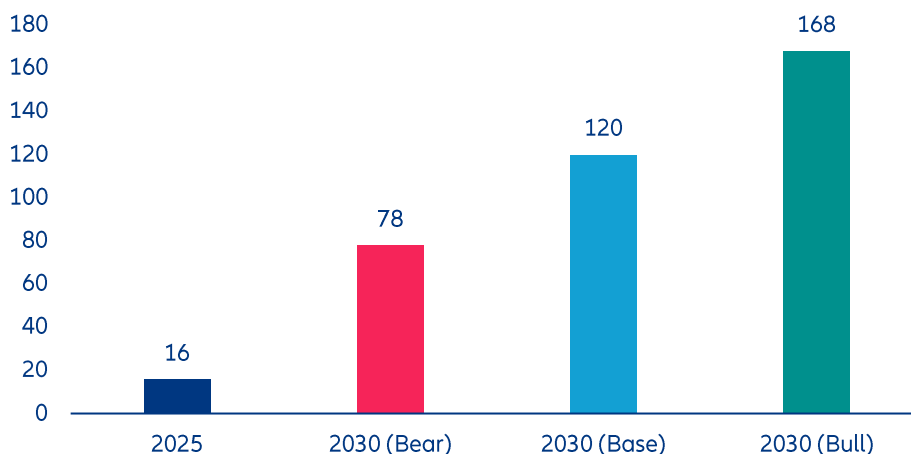
Table 2: Business applications of LEO satellites services

Application	What LEO unlocks
Rural & enterprise broadband	Fiber-like speed/latency where terrestrial build-out is partial
Direct-to-cell (sat-to-smartphone)	Unmodified phones connect in dead zones, erasing mobile not-spots
Mobile/5G backhaul	Connects cell towers and extends 5G where fiber can't reach
IoT / asset tracking / M2M	Ubiquitous low-power links for logistics, agriculture, energy, supply chains
Earth observation & geospatial analytics	High-resolution imagery for climate, urban planning, insurance, defense ISR
Maritime & aviation connectivity	Low-latency in-flight Wi-Fi and vessel comms
Resilience & sovereign/defense comms	Backup connectivity in case of climate events / secure government/military links
Latency-sensitive niches	Use cases GEO couldn't serve

Source: Allianz Research

Demand sizing: from niche to infrastructure. The high expansion potential of the LEO satellite market relies on its own characteristics. Small footprints mandate thousands of satellites, short lifespans make replacement recurring and low unit costs make that perpetual replenishment pay – a self-sustaining demand engine, not a one-and-done build. The services market was valued at approximately USD16bn in 2025. Current forecasts project growth to USD120bn by 2030, a 7-8x expansion in under a decade, driven primarily by enterprise and industrial demand rather than consumer broadband. This sits within a broader space economy currently valued at around USD600bn, which is projected to triple to over USD1.8trn by 2035, with LEO services and defense-related space infrastructure displaying the highest growth rates.

Figure 2: Estimated revenue scenario of LEO satellites services market by 2030



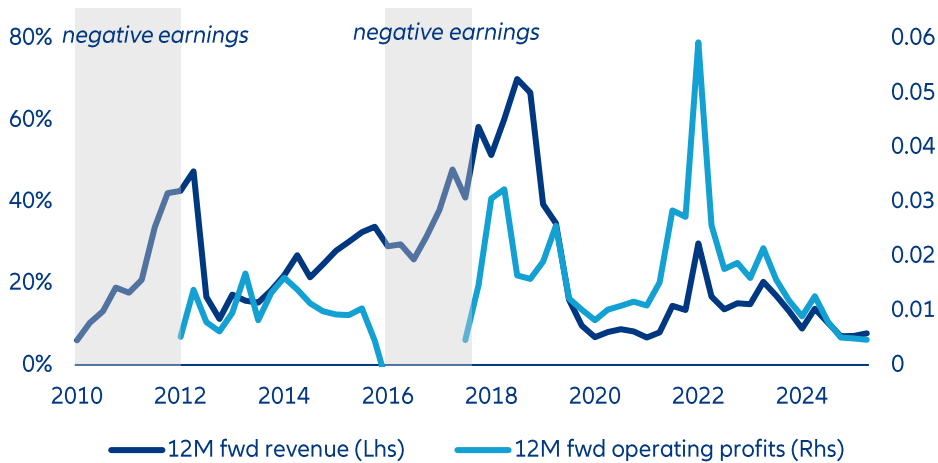
Source: Allianz Research

The geopolitical dimension compounds this demand. The drone strikes on three Amazon-linked data center facilities in the Gulf region in early 2026, and the resulting pause in data center investment decisions across GCC markets, illustrate a structural shift: physical terrestrial infrastructure is increasingly exposed to conflict risk. Distributed LEO constellations, by contrast, present no single point of failure and no fixed geographic target. For enterprises and governments operating in geopolitically unstable regions, satellite connectivity is becoming a resilience requirement, not merely a coverage solution.

The US is well ahead in the race, but China is running hard

SpaceX and the IPO as a market signal. Starlink, SpaceX's satellite internet division, has no comparable competitor today. With over 9,000 satellites already in orbit and regulatory approval filed for up to one million, it operates at a scale that compresses unit economics in ways no rival can currently replicate. The vertical integration is equally important: SpaceX designs the satellites, manufactures them at scale, launches them on its own Falcon 9 and Starship vehicles and operates the ground network. This eliminates the margin leakage that plagues operators dependent on third-party launch providers and means that SpaceX can deploy, replace and upgrade its constellation at a cost structure competitors cannot match. Its IPO converted this operational dominance into a public market signal. At a listing valuation of USD1.75trn, Starlink's implied revenue multiple is aggressive, reflecting not just current commercial contracts but the embedded option on becoming the connectivity backbone for the global AI economy. For investors, the relevant question is not whether SpaceX is worth USD1.75trn or so but what the IPO reveals about where capital is flowing and what it is pricing in. IPO valuations are rarely a reflection of current fundamentals; they are a measure of investors' conviction in a company's ability to dominate future markets. Tesla proved that markets can assign extraordinary value to a perceived pioneer long before revenues and profits justify it, pricing in opportunities from robotaxis, humanoids and other disruptive technologies. Would SpaceX benefit from the same premium? Perhaps as the group is the leader in an industry in the early phase of development. Yet the higher the expectations, the narrower the margin for error. When valuation is built on future promises rather than present fundamentals, execution risk becomes the ultimate determinant of success.

Figure 3: Tesla case study – historical disconnection between market value and fundamentals is at its climax



Source: Allianz Research

Amazon's Project Kuiper is the most significant near-term competitive development. With approximately 3,200 satellites authorized and the first commercial launches beginning in 2026 (carried in part by Arianespace's Ariane 6), Kuiper is not a standalone business. It is Amazon Web Services' hedge against dependence on SpaceX for the edge connectivity that enterprise cloud customers will increasingly require. Amazon has already invested over USD10bn in Kuiper development. The strategic logic is identical to the reasoning behind AWS itself in 2006: build the infrastructure that your customers will need before they are forced to buy it from a competitor. This matters for the competitive landscape because Kuiper's primary market is not consumer broadband, it is enterprise and government connectivity, the same market where Starlink is making its most aggressive commercial moves. Two well-capitalized, vertically-oriented operators competing for the same enterprise segment will compress service pricing over the medium term, which benefits end users but narrows the commercial moat for pure-play satellite service operators.

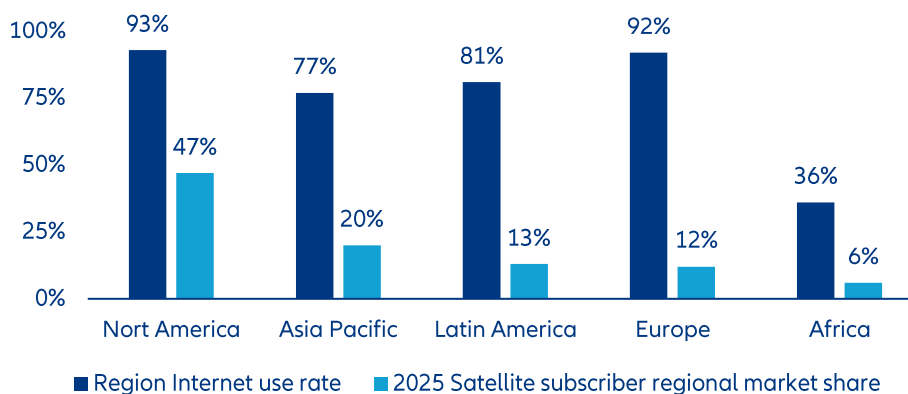
China's satellite ambitions are substantial and systematically underappreciated in Western market analysis. China SatNet, the state-backed operator, has filed for 13,000 satellites, Shanghai Spacecom Satellite Technology (SSST) has filed for a further 15,000. A broader accounting of filed Chinese constellations - against less than 1000 currently in orbit - puts the pipeline around 30,000 satellites across multiple operators, though execution timelines vary widely. The Chinese program differs from US operators in its strategic logic. Where SpaceX and Kuiper are primarily commercial ventures with sovereign dimensions, China's LEO build-out is sovereign infrastructure with commercial dimensions. State funding removes the capital constraint that limits most commercial competitors, and the domestic manufacturing base, now extending to satellite bus production at mass scale, replicates the cost-compression playbook that made China competitive in solar panels and electric vehicles. Spectrum and orbital slot allocation, governed under ITU first-come-first-served rules, creates an additional urgency: slots occupied by Chinese constellations are slots unavailable to future US or European entrants. The net result is an emerging bipolar structure in LEO: US commercial operators with a current lead, and Chinese state-backed operators with the capital and political mandate to close the gap regardless of short-term economics. Every other actor is operating in the margin between these two gravitational forces.

Emerging markets are the battleground

The case for LEO satellites is strongest where terrestrial infrastructure is weakest. Fixed broadband penetration remains below 40% in Africa and at or near 80% threshold in Asia Pacific and Latin America regions - against over 90% in both Europe and North America - with some noticeable discrepancy between large and small countries. Rural populations in these regions are not waiting for fiber rollout, they are waiting for connectivity that may never arrive via terrestrial means within a commercially viable timeframe. The infrastructure economics simply do not

close: low population density, difficult terrain and limited purchasing power make laying cable or building cell towers unattractive for private operators and fiscally difficult for governments. LEO satellites change the economics in the same way mobile phones did in the 2000s. A single orbital shell provides blanket coverage regardless of ground infrastructure, at costs that are declining rapidly as constellation scale increases and with it the range of services provided based on market structure and demand. The applications that follow (e.g. precision agriculture, remote diagnostics, e-government services, financial inclusion, supply-chain connectivity for commodity industries etc.) could be structural economic enablers for economies where the productivity gap between connected and unconnected populations is large. Emerging markets are therefore a large untapped demand pool. The question is not whether these economies will depend on LEO satellite connectivity but whose infrastructure they will depend on, and on whose terms.

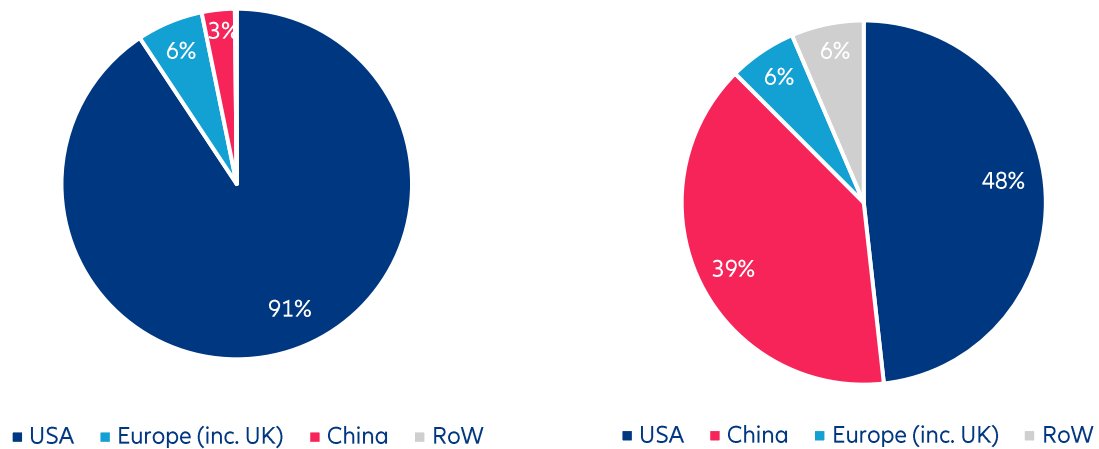
Figure 4: Internet user rate (as % of total population) and regional market share of internet broadband services via satellites



Sources: ITU DataHub, Allianz Research

China understood this dynamic earlier and more strategically than its Western competitors. Chinese state-backed satellite operators are being explicitly positioned as the satellite connectivity layer of the Belt and Road Initiative's digital extension and expect to substantially ramp-up their position in that segment, which has so far been broadly given to US firms. Governments across Africa, the Middle East and Southeast Asia that have accepted BRI infrastructure financing are natural targets for preferential access to Chinese LEO services, bundled with ground station equipment, network management systems and data-handling arrangements that create lasting technical and political dependency. The model replicates the playbook used with Huawei 5G equipment: offer below-market pricing backed by a state subsidy, establish technical standards and integration points that are difficult to unwind and convert infrastructure dependency into geopolitical leverage over time. In the satellite context, the dependency runs deeper – a government whose critical communications infrastructure, agricultural monitoring systems and government data flows run through a foreign-operated constellation has limited practical ability to renegotiate terms after the fact. This trajectory already visible in the terrestrial digital infrastructure decisions made across BRI partner economies over the past decade, now extending into orbital infrastructure to keep control over the whole technology on which its AI infrastructure is based. Accounting for almost 40% of LEO satellites to become effective by 2030, China displays a double ambition : 1) reducing external technology reliance and 2) increasing its digital footprint over high-growth-potential data. For Western policymakers and investors assessing the geopolitical architecture of the LEO race, the emerging market connectivity gap is a key arena where the long-term balance of power in global digital infrastructure will be decided.

Figure 5: Current and expected regional distribution of LEO satellites fleet



Sources: Corporate public announcement, Allianz Research

Not all emerging economies are passive recipients. A handful of larger markets have the domestic capability, market size and political will to negotiate terms or develop partial autonomy. India is the most consequential case. The Indian Space Research Organization has established commercial launch credentials and is actively developing indigenous satellite constellation capabilities. The Indian government has so far blocked or heavily conditioned Starlink's market access, pending domestic regulatory frameworks that ensure data localization and sovereign oversight. With over 1.4bn people and a rapidly expanding digital economy, India's connectivity market is large enough that operators need it more than it needs any single operator, a negotiating position no smaller emerging economy can replicate. Brazil holds a structural geographic advantage that gives it disproportionate leverage in the launch segment: equatorial launch sites reduce the fuel cost of reaching LEO by a material margin, and Brazil's Alcântara launch base is among the best-positioned in the world. This asset has been underexploited. Although a 2019 technology safeguards agreement with the US opened the facility to US commercial operators, launches remain low. But as launch economics tighten and competition intensifies, Alcântara could become a genuine strategic asset. The UAE and Saudi Arabia are pursuing sovereign space capabilities – including domestic satellite manufacturing and launch ambitions – as part of broader economic diversification programs. Their capital depth and tolerance for long investment horizons makes them credible entrants in the sovereign LEO infrastructure segment. However, for the remaining 150-plus economies without the scale, capital or technical base to negotiate on equal terms, the LEO connectivity future is one of dependence: on US commercial operators, on Chinese state-backed alternatives or on the institutional frameworks that multilateral bodies like the ITU and the UN Office for Outer Space Affairs can realistically enforce. None of those options guarantees equitable access or protects against infrastructure leverage.

Europe's narrow window: Specialize or be left behind

Europe is not absent from the LEO landscape. But its position reflects a pattern familiar from the cloud development era: strong institutional intent, inadequate commercial execution. Two programs define Europe's current exposure. Eutelsat OneWeb, formed from the merger of French operator Eutelsat and UK-based OneWeb, operates a constellation of approximately 650 LEO satellites and serves enterprise and government connectivity markets. It is the only European operator with a functioning LEO constellation at meaningful scale, but its financial position is constrained (the merger was partly a rescue operation) and as a result its business strategy is more qualitative than quantitative-centric (440 satellites to be deployed by 2027-2030, mostly replacements of old models) compared with US or Chinese competitors. IRIS², the pan-European institutional constellation backed by the EU and European Space Agency, targets 280 satellites (260 LEO, 20 GEO) with a 2030 operational target. It is designed primarily for government and critical infrastructure connectivity, offering encrypted communications resilient to conflict disruption. Eleven companies form the industrial consortium. IRIS² is a genuine strategic achievement, a coordinated European response to the sovereignty vulnerabilities exposed by dependence on non-European satellite operators, but its scale is deliberately limited and its commercial ambitions are constrained by its sovereign mandate. Against a US-China combined pipeline running well above 100,000 satellites, Europe's credible commercial constellation capacity – including replacement of old models – should remain below 1,000 level. No European operator has the capital depth of Blue Origin, the launch verticality of SpaceX or the hyperscaler integration of Amazon Kuiper.

The ASML analogy: own the chokepoint. The history of European industrial competitiveness in technology-intensive sectors suggests a different model: identify a structural bottleneck in the value chain, invest deeply in the capabilities required to dominate it and leverage that position to capture value regardless of which end-product competitors win. ASML's dominance in extreme UV lithography is the benchmark case. ASML does not compete in the semiconductor end-market, it supplies the tool that every leading chipmaker requires to produce advanced semiconductors. Its market position is not the result of having the largest chip factory, it is the result of owning an irreplaceable step in the production process. Arianespace occupies an analogous position in launch operations. Europe's launch infrastructure, Ariane 6, with the Vega-C small launcher alongside, is not the cheapest or remains constrained by its non-reusable model, but it is an independent, reliable, sovereign-controlled capability. Critically, Amazon has contracted Arianespace to launch a significant tranche of Kuiper satellites on Ariane 6. This is not a minor commercial deal: it is validation that a US hyperscaler investing in its own constellation has chosen European launch capacity as a component of its supply chain. But with around 10 flights per year compared with over 100 for its US rivals SpaceX, Europe is today not yet really sized to compete with the US and China, at least on the commercial segment. Next-generation initiatives point to a credible response: MaiaSpace, an ArianeGroup subsidiary, is developing the partially reusable Maia mini-launcher (maiden flight expected from 2026–27), drawing on the Prometheus engine and the Themis reusable first-stage demonstrator, whose first hop test is slated for 2026. These building blocks are intended to feed a reusable first stage for Ariane Next in the 2030s, marking the path by which Europe could close the reusability and cost gap and consolidate its position across the LEO supply chain. The commercial proof-of-concept for the “ASML model” already exists. Rather than attempting to build a commercial constellation at US or Chinese scale – an effort that would require capital and political coordination that is not available – European industrial policy should concentrate on making Arianespace's launch capability indispensable across operators regardless of nationality, in the same way that ASML's tools are used by the world's leading chip manufacturers.

Table 3: key technical differences between main LEO satellite launchers

Criteria	Arianespace	SpaceX	Blue Origin	United Launch Alliance
LEO payload	22 t	23 t Falcon 9 / >100 t Starship	45 t	27 t+
Reusable	No	Yes	Yes	No
Fairing volume	Large	Moderate	Very large	Large
Current cadence	Low (~5-10/yr)	Very high (>100/yr)	Very low	Moderate
Constellation experience	OneWeb, Kuiper	Starlink leader	Kuiper-focused	Kuiper
Sovereign access	European	US	US	US
Vertical integration	No	Very strong	Strong	No
Cost competitiveness	Medium	Excellent	Potentially excellent	Weak

Source: Allianz Research

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